

[ACJC/2024/Q1]

1 Name your Jupyter Notebook as:

TASK1_<your name>_<centre number>_<index number>.ipynb

In this task, you are required to implement a sorting algorithm known as Stalin Sort.

The algorithm runs as follows:

1. Starting from the beginning of the list, compare each item with the one following it in the list.
2. If the i^{th} item in the list is **not** in the correct order relative to the $(i + 1)^{\text{th}}$ item, pop the $(i + 1)^{\text{th}}$ item from the list and insert it at the beginning of the list.
3. Move on to compare the $(i + 1)^{\text{th}}$ item with the $(i + 2)^{\text{th}}$ item.
4. Continue this process until you reach the end of the list.
5. Steps 1 to 4 are considered a pass. Repeat steps 1 to 4 until the list is sorted. The list is determined to be sorted when no items are inserted at the front of the list during a complete pass.

The following example shows how the algorithm would sort a list of integers in ascending order.

Pass 0: [2, 0, 1, 4, 3]

Pass 1: [3, 1, 0, 2, 4]

Pass 2: [2, 0, 1, 3, 4]

Pass 3: [1, 0, 2, 3, 4]

Pass 4: [0, 1, 2, 3, 4]

For each of the sub-tasks, add a comment statement at the beginning of the code, using the hash symbol '#' to indicate the sub-task the program code belongs to, for example:

```
In [1]: #Task 1.1
        Program code
        Output:
```

Task 1.1

The value of a string is defined as the number of characters in the string multiplied by the number of vowels in the string.

Write a function `string_value(s1)` that returns the value of a string. Note that the string might contain a mixture of upper- and lowercase characters. [3]

Task 1.2

Write a function `StalinSort(lst)` that would sort a list of strings in ascending order of value, as determined in Task 1.1. If two strings have the same value, sort them in lexicographic order.

Test your function with the following test cases:

```
print(StalinSort(["Peter", "Josh", "See", "Hi" ]))
print(StalinSort(["PetEr", "JOsh", "Hi", "See" ]))
print(StalinSort(["Peter", "Hello", "BYE", "Hi", "ah", "bbbb" ])) [8]
```

Task 1.3

Write program code to:

- read the text file `words.txt` and store the content in a list;
- sort the list using `StalinSort(lst)`;
- display the sorted list.

[3]

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Task 1.4

Write a function `QuickSort(lst)` that would sort a list of strings in ascending order of value, as determined in Task 1.1, using a quicksort algorithm. For strings of the same value, sort them in alphabetical order. [7]

Task 1.5

Write a function `BinarySearch(lst, value)` that will perform a binary search through the sorted list of words from Task 1.3 for any word whose value, as determined in Task 1.1, is `value`.

- If such a word can be found, return its index in the list.
- If such a word cannot be found, return `False`.

[4]

Save your Jupyter Notebook for Task 1.